

INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

CHAPTER 7: References

by

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7 References

- [1] Schär, F. (2021). Decentralized finance: On blockchain- and smart contract-based financial markets. *Federal Reserve Bank of St. Louis Review*, 103(2), 153–174. doi:10.20955/r.103.153-74
- [2] Packin, N. G., & Lev-Aretz, Y. (2024). Decentralized credit scoring: Black box 3.0. *American Business Law Journal*, 61(2), 91–111. doi:10.1111/ablj.12240
- [3] Do, T., Do, D., & Nguyen, L. (2023). On-chain reputation ranking by Adaptive Weighted PageRank. In *2023 RIVF International Conference on Computing and Communication Technologies (RIVF)* (pp. 318–323). IEEE. doi:10.1109/RIVF60135.2023.10471809
- [4] Do, H., & Do, T. (2023). PageRank and HodgeRank on Ethereum transactions: A measure for social credit. *International Journal of Software Innovation*, 11(1), 1–13. doi:10.4018/IJSI.315737
- [5] Nakamoto, S. (2008). *Bitcoin: A peer-to-peer electronic cash system*. bitcoin.org/bitcoin.pdf
- [6] Szabo, N. (1997). Formalizing and securing relationships on public networks. *First Monday*, 2(9). doi:10.5210/fm.v2i9.548
- [7] Buterin, V. (2014). *A next-generation smart contract and decentralized application platform*. Ethereum Foundation.
- [8] Werner, S., Perez, D., Gudgeon, L., Klages-Mundt, A., Harz, D., & Knottenbelt, W. (2022). SoK: Decentralized finance (DeFi). In *Proceedings of the 4th ACM Conference on Advances in Financial Technologies* (pp. 30–46). ACM. doi:10.1145/3558535.3559780
- [9] Gudgeon, L., Perez, D., Harz, D., Livshits, B., & Gervais, A. (2020). The decentralized financial crisis. In *2020 Crypto Valley Conference on Blockchain Technology* (pp. 1–15). IEEE.

- [10] Harvey, C. R., Ramachandran, A., & Santoro, J. (2021). *DeFi and the future of finance*. Wiley.
- [11] Douceur, J. R. (2002). The Sybil attack. In *Peer-to-Peer Systems* (pp. 251–260). Springer. doi:10.1007/3-540-45748-8_24
- [12] Lin, D., Wu, J., Fu, Q., Zheng, Z., & Chen, T. (2025). RiskProp: Account risk rating on Ethereum via de-anonymous score and network propagation. *IEEE Transactions on Dependable and Secure Computing*, 22(3), 2132–2145. doi:10.1109/TDSC.2024.3475825
- [13] Hassija, V., Bansal, G., Chamola, V., Kumar, N., & Guizani, M. (2020). Secure lending: Blockchain and prospect theory-based decentralized credit scoring model. *IEEE Transactions on Network Science and Engineering*, 7(4), 2566–2575. doi:10.1109/TNSE.2020.2982488
- [14] Kotb, M. O. (2023). Credit scoring using machine learning algorithms and blockchain technology. In *2023 Intelligent Methods, Systems, and Applications (IMSA)* (pp. 381–386). IEEE. doi:10.1109/IMSA58542.2023.10217411
- [15] Page, L., Brin, S., Motwani, R., & Winograd, T. (1998). *The PageRank citation ranking: Bringing order to the web* (Technical Report). Stanford Digital Library Technologies Project.
- [16] Kamvar, S. D., Schlosser, M. T., & Garcia-Molina, H. (2003). The EigenTrust algorithm for reputation management in P2P networks. In *Proceedings of the 12th International Conference on World Wide Web* (pp. 640–651). ACM. doi:10.1145/775152.775242
- [17] Gyöngyi, Z., Garcia-Molina, H., & Pedersen, J. (2004). Combating web spam with TrustRank. In *Proceedings of the Thirtieth International Conference on Very Large Data Bases* (pp. 576–587). VLDB Endowment.
- [18] Ethereum Improvement Proposals. (2015). ERC-20 token standard (EIP-20). EIP-20
- [19] Ethereum Improvement Proposals. (2020). ERC-20 permit extension (EIP-2612). EIP-2612
- [20] Spearman, C. (1904). The proof and measurement of association between two things. *The American Journal of Psychology*, 15(1), 72–101. doi:10.2307/1412159
- [21] Kendall, M. G. (1938). A new measure of rank correlation. *Biometrika*, 30(1/2), 81–93.

doi:10.2307/2332226

- [22] Cronbach, L. J., & Meehl, P. E. (1955). Construct validity in psychological tests. *Psychological Bulletin*, 52(4), 281–302. doi:10.1037/h0040957
- [23] Langville, A. N., & Meyer, C. D. (2006). *Google's PageRank and beyond: The science of search engine rankings*. Princeton University Press.
- [24] Wu, J., Yuan, Q., Lin, D., You, W., Chen, W., Chen, C., & Zheng, Z. (2022). Who are the phishers? Phishing scam detection on Ethereum via network embedding. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 52(2), 1156–1166. doi:10.1109/TSMC.2020.3016821
- [25] Bellini, E., Iraqi, Y., & Damiani, E. (2020). Blockchain-based distributed trust and reputation management systems: A survey. *IEEE Access*, 8, 21127–21151. doi:10.1109/ACCESS.2020.2969820
- [26] Almasoud, A. S., Hussain, F. K., & Hussain, O. K. (2020). Smart contracts for blockchain-based reputation systems: A systematic literature review. *Journal of Network and Computer Applications*, 170, 102814. doi:10.1016/j.jnca.2020.102814
- [27] Zhou, L., Xiong, X., Ernstberger, J., Chaliasos, S., Wang, Z., Wang, Y., Qin, K., Wattenhofer, R., Song, D., & Gervais, A. (2023). SoK: Decentralized finance (DeFi) attacks. In *2023 IEEE Symposium on Security and Privacy* (pp. 2444–2461). IEEE. doi:10.1109/SP46215.2023.10179435
- [28] Victor, F., & Weintraud, A. M. (2021). Detecting and quantifying wash trading on decentralized cryptocurrency exchanges. In *Proceedings of the Web Conference 2021* (pp. 23–32). ACM. doi:10.1145/3442381.3449824
- [29] Cong, L. W., Li, X., Tang, K., & Yang, Y. (2023). Crypto wash trading. *Management Science*, 69(11), 6427–6454. doi:10.1287/mnsc.2021.02709
- [30] Qin, K., Zhou, L., & Gervais, A. (2022). Quantifying blockchain extractable value: How dark is the forest? In *2022 IEEE Symposium on Security and Privacy* (pp. 198–214). IEEE. doi:10.1109/SP46214.2022.9833734

- [31] Xia, P., Wang, H., Gao, B., Su, W., Yu, Z., Luo, X., Zhang, C., Xiao, X., & Xu, G. (2021). Trade or trick? Detecting and characterizing scam tokens on Uniswap decentralized exchange. *Proceedings of the ACM on Measurement and Analysis of Computing Systems*, 5(3), 1–26. doi:10.1145/3491051
- [32] Luo, J., Kang, H., Zheng, S., & Liu, X. (2025). Toward resilient airdrop mechanisms: Empirical measurement of hunter profits and airdrop game theory modeling. In *2025 IEEE International Conference on Blockchain and Cryptocurrency (ICBC)* (pp. 1–5). IEEE. doi:10.1109/ICBC64466.2025.11114597
- [33] Messick, S. (1995). Validity of psychological assessment: Validation of inferences from persons' responses and performances as scientific inquiry into score meaning. *American Psychologist*, 50(9), 741–749. doi:10.1037/0003-066X.50.9.741
- [34] Efron, B. (1979). Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1), 1–26. doi:10.1214/aos/1176344552
- [35] DiCiccio, T. J., & Efron, B. (1996). Bootstrap confidence intervals. *Statistical Science*, 11(3), 189–228. doi:10.1214/ss/1032280214
- [36] Wood, G. (2014). *Ethereum: A secure decentralised generalised transaction ledger* (Ethereum Yellow Paper). Ethereum Project.
- [37] Kitzler, S., Victor, F., Saggese, P., & Haslhofer, B. (2023). Disentangling decentralized finance (DeFi) compositions. *ACM Transactions on the Web*, 17(2), 1–26. doi:10.1145/3532857
- [38] Qin, K., Zhou, L., Livshits, B., & Gervais, A. (2021). Attacking the DeFi ecosystem with flash loans for fun and profit. In *Financial Cryptography and Data Security* (pp. 3–25). Springer. doi:10.1007/978-3-662-64322-8_1
- [39] Angeris, G., & Chitra, T. (2020). Improved price oracles: Constant function market makers. In *Proceedings of the 2nd ACM Conference on Advances in Financial Technologies* (pp. 80–91). ACM. doi:10.1145/3419614.3423251

- [40] Victor, F. (2020). Address clustering heuristics for Ethereum. In *Financial Cryptography and Data Security. FC 2020* (pp. 617–633). Springer. doi:10.1007/978-3-030-51280-4_33
- [41] Bartoletti, M., Chiang, J. H., & Lluch-Lafuente, A. (2021). SoK: Lending pools in decentralized finance. In *Financial Cryptography and Data Security. FC 2021 International Workshops* (pp. 553–578). Springer. doi:10.1007/978-3-662-63958-0_40
- [42] Gervais, A., Karame, G. O., Wüst, K., Glykantzis, V., Ritzdorf, H., & Capkun, S. (2016). On the security and performance of proof of work blockchains. In *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security* (pp. 3–16). ACM. doi:10.1145/2976749.2978341
- [43] Gudgeon, L., Moreno-Sanchez, P., Roos, S., McCorry, P., & Gervais, A. (2020). SoK: Layer-two blockchain protocols. In *Financial Cryptography and Data Security* (pp. 201–226). Springer. doi:10.1007/978-3-030-51280-4_12
- [44] Thibault, L. T., Sarry, T., & Hafid, A. S. (2022). Blockchain scaling using rollups: A comprehensive survey. *IEEE Access*, 10, 93039–93054. doi:10.1109/ACCESS.2022.3200051
- [45] Kalodner, H., Goldfeder, S., Chen, X., Weinberg, S. M., & Felten, E. W. (2018). Arbitrum: Scalable, private smart contracts. In *27th USENIX Security Symposium* (pp. 1353–1370). USENIX Association.
- [46] GMX. (n.d.). *GMX V2 documentation*. docs.gmx.io
- [47] Akerer, D., Hugonnier, J., & Jermann, U. (2024). *Perpetual futures pricing* (NBER Working Paper No. 32936). National Bureau of Economic Research. doi:10.3386/w32936
- [48] Liu, X. F., Jiang, X.-J., Liu, S.-H., & Tse, C. K. (2021). Knowledge discovery in cryptocurrency transactions: A survey. *IEEE Access*, 9, 37229–37254. doi:10.1109/ACCESS.2021.3062652
- [49] Leskovec, J., Rajaraman, A., & Ullman, J. D. (2020). *Mining of massive datasets* (3rd ed.). Cambridge University Press.
- [50] Kleinberg, J. M. (1999). Authoritative sources in a hyperlinked environment. *Journal of the*

ACM, 46(5), 604–632. doi:10.1145/324133.324140

- [51] Li, W., Bu, J., Li, X., & Chen, X. (2022). Security analysis of DeFi: Vulnerabilities, attacks and advances. In *2022 IEEE International Conference on Blockchain* (pp. 488–493). IEEE.
- [52] Perez, D., Werner, S. M., Xu, J., & Livshits, B. (2021). Liquidations: DeFi on a knife-edge. In *Financial Cryptography and Data Security* (pp. 457–476). Springer.
- [53] Klages-Mundt, A., Harz, D., Gudgeon, L., Liu, J.-Y., & Minca, A. (2020). Stablecoins 2.0: Economic foundations and risk-based models. In *Proceedings of the 2nd ACM Conference on Advances in Financial Technologies* (pp. 59–79). ACM.
- [54] Carter, N., & Jeng, L. (2021). DeFi protocol risks: The paradox of DeFi. In *Regtech, Suptech and Beyond: Innovation and Technology in Financial Services*. Risk Books.
- [55] Daian, P., Goldfeder, S., Kell, T., Li, Y., Zhao, X., Bentov, I., Breidenbach, L., & Juels, A. (2020). Flash Boys 2.0: Frontrunning in decentralized exchanges, miner extractable value, and consensus instability. In *2020 IEEE Symposium on Security and Privacy* (pp. 910–927). IEEE. doi:10.1109/SP40000.2020.00040
- [56] Makarov, I., & Schoar, A. (2020). Trading and arbitrage in cryptocurrency markets. *Journal of Financial Economics*, 135(2), 293–319. doi:10.1016/j.jfineco.2019.07.001
- [57] Resnick, P., & Zeckhauser, R. (2002). Trust among strangers in Internet transactions: Empirical analysis of eBay’s reputation system. In *The Economics of the Internet and E-commerce* (pp. 127–157). Emerald.
- [58] Haveliwala, T. H. (2003). Topic-sensitive PageRank: A context-sensitive ranking algorithm for web search. *IEEE Transactions on Knowledge and Data Engineering*, 15(4), 784–796. doi:10.1109/TKDE.2003.1208999
- [59] Jeh, G., & Widom, J. (2003). Scaling personalized web search. In *Proceedings of the 12th International Conference on World Wide Web* (pp. 271–279). ACM. doi:10.1145/775152.775191
- [60] Antonopoulos, A. M., & Wood, G. (2018). *Mastering Ethereum: Building smart contracts and*

DApps. O'Reilly Media.

- [61] Narayanan, A., Bonneau, J., Felten, E., Miller, A., & Goldfeder, S. (2016). *Bitcoin and cryptocurrency technologies*. Princeton University Press.
- [62] Torres, C. F., Steichen, M., & State, R. (2019). The art of the scam: Demystifying honeypots in Ethereum smart contracts. In *28th USENIX Security Symposium* (pp. 1591–1607). USENIX Association.
- [63] Zetsche, D. A., Arner, D. W., & Buckley, R. P. (2020). Decentralized finance. *Journal of Financial Regulation*, 6(2), 172–203. doi:10.1093/jfr/fjaa010
- [64] Halu, A., Mondragón, R. J., Panzarasa, P., & Bianconi, G. (2013). Multiplex PageRank. *PLoS ONE*, 8(10), e78293. doi:10.1371/journal.pone.0078293
- [65] De Domenico, M., Solé-Ribalta, A., Omodei, E., Gómez, S., & Arenas, A. (2015). Ranking in interconnected multilayer networks reveals versatile nodes. *Nature Communications*, 6, 6868. doi:10.1038/ncomms7868